SoftWhere: Differentiable Selection Done Right

A sampling-free, multi-foveal TokenLearner selector for LookWhere — proposal with preliminary de-risking results

1. Background and Motivation

- **LookWhere** trains its selector by distilling the DINOv2 teacher's last-layer attention into a 2D importance map, then takes a hard top-k. Selection is non-differentiable, so the extractor's loss does not flow back to *where* to look.
- The authors tried to make the selection differentiable end-to-end by testing Gumbel-top-k (with a straight-through estimator) and REINFORCE (Table 12, Fig. 7b). Both of these methods hurt performance, and plain attention distillation won. They give a precise reason for this: training fails to converge without the map-distillation loss, which is a KL regularizer toward the teacher's "policy." All of their differentiable configurations already included that regularizer, but the extractor-signal gradient on top of it still did not help.
- **TokenLearner** calculates soft spatial attention in a fully differentiable way that avoids sampling altogether. Instead of relying on Gumbel or action sampling, it generates S distinct attention maps, meaning every token focuses on a different area.

2. The Gap and the Bar to Beat

The honest distinction is **sampling-based vs. sampling-free**, not “high-variance vs. low-variance.” LookWhere's Gumbel-top-k is already the low-variance reparameterized option used to avoid REINFORCE's variance, and it still failed (it was the worst at k=16/72). So variance alone is not the explanation. What both their methods share is **stochastic sampling of the selection** (plus, for Gumbel-ST, straight-through bias on the selection distribution). A TokenLearner selector removes the sampling entirely while keeping the mandatory KL regularizer. Because even the low-variance Gumbel-ST failed, a clean result here is itself diagnostic.

Baselines (LookWhere Table 12; ViT-S, kNN top-1; cls = ImageNet-HR class token, seg = ADE20K patch tokens). SoftWhere must clear the bolded row:

Selector-map training	cls k=16	cls k=72	cls k=128	seg k=16	seg k=72	seg k=128
none (stop-grad distill) — winner	**50.9**	**63.0**	**66.5**	**51.1**	**60.2**	**62.9**

Gumbel-top-k (straight-through)	44.5	61.4	66.2	49.1	58.9	62.5
REINFORCE	47.9	61.8	66.0	50.7	59.6	62.8

Note the gaps **collapse at k=128** (66.5/66.2/66.0; 62.9/62.8) — the clearest room to win is at k=16 and k=72.

3. Hypothesis

A sampling-free TokenLearner-SR selector can match or exceed LookWhere when it preserves three properties: teacher-aligned map distillation, resolution parity with the LookWhere selector grid, and a deterministic selection policy that keeps the S foveal maps separate until patch selection. The updated hypothesis is therefore more specific than “soft selection is enough”: SoftWhere should improve multi-object coverage when four high-resolution foveal maps are selected per fovea with light spatial non-maximum suppression, while retaining the KL map regularizer LookWhere found essential.

The core scientific test remains direct: if SoftWhere beats LookWhere under the same kNN classification and segmentation protocols, then the failure mode in LookWhere’s Gumbel-top-k and REINFORCE trials was not differentiability itself. If it ties or loses, the coverage diagnostics still isolate resolution, selection policy, and extractor-gradient effects.

4. Method

Selector. Replace LookWhere’s per-patch MLP SelectorHead with TokenLearner-SR: a TokenLearner head that emits S high-resolution foveal scoring maps using the same super-resolution idea needed for parity with the LookWhere selector map. The current anchor setting is v10, S=4, convolutional super-resolution, and no explicit diversity regularizer.

Selection / fetch. The main positive policy is per-map NMS: allocate the patch budget across the foveal maps, select high-scoring patches from each map, suppress nearby redundant selections, and union the selected high-resolution patches. Aggregate top-k, per-map top-k without NMS, and distance-penalized aggregate selection remain ablations.

Training. Stage 1 distills the TokenLearner-SR maps to the frozen LookWhere-DINOv2 teacher with the same KL-style map regularizer. Stage 2 adds extractor-feature signal end-to-end while retaining map regularization. The mini diagnostic used lambda_cls=1, lambda_map=1, lambda_pat=0, and lambda_div=0, initialized from the best distilled div0 head.

Evaluation. Coverage diagnostics use ADE20K small multi-object recall at k={16,72,128,136}, plus teacher agreement, IoU, map overlap, and qualitative foveal-map visualization. The final proposal experiments still need LookWhere-comparable kNN classification on ImageNet-HR and kNN segmentation on ADE20K patch tokens.

5. Updated Preliminary Results

We implemented SoftWhere against the public LookWhere-DINOv2 checkpoint and ran seven diagnostics recorded in softwhere/personal/results. The results changed the proposal from a broad TokenLearner idea into a narrower, better-supported method: TokenLearner-SR plus per-fovea selection, with mini end-to-end classifier-signal training as the current best head.

5.1 Resolution parity fixes the early coverage failure

The earliest low-resolution TokenLearner maps were too coarse for small ADE20K objects. After adding TokenLearner-SR with convolutional super-resolution, the v10/S=4/div1 head learned the LookWhere saliency in minutes: distillation loss fell from 1.098 to 0.057, teacher top-k recall was 0.495, teacher IoU was 0.334, and random recall was only 0.102. The foveal maps were also nearly non-overlapping in that run, with map overlap 0.001.

Selector at k=136	ADE20K object recall
LookWhere single map	0.701
SoftWhere aggregate	0.540
SoftWhere multi-foveal top-k/S	0.674
Random	0.606

This result overturned the initial negative spike. SoftWhere was no longer below random; it was above random and close to LookWhere, but not yet ahead.

5.2 Selection policy is decisive

Holding the v10/SR/div1 head fixed, the policy ablation showed that the foveal structure is useful only if selection preserves it. A soft aggregate loses the coverage signal, while per-map NMS turns the same maps into the first positive SoftWhere coverage result.

Policy	Object recall
LookWhere single map	0.7012
SoftWhere aggregate top-k	0.5404
SoftWhere top-k/S per fovea	0.6738
SoftWhere per-map NMS	0.7389
SoftWhere aggregate + distance penalty	0.6902
Random	0.6055

Per-map NMS beats LookWhere by +0.0377 absolute recall at k=136. The distance-penalized aggregate does not beat LookWhere, suggesting the win is not merely “spread patches out”; it depends on retaining per-fovea structure.

5.3 Robustness sweeps identify the main distilled configuration

Sweeping NMS distance across k={16,72,128,136} showed nms_dist=2 as the sweet spot. With the div1 head, SoftWhere beats LookWhere at k=72, k=128, and k=136, but still trails at k=16.

k	Best div1 SoftWhere	LookWhere	Margin
16	0.2406	0.2622	-0.0216
72	0.5817	0.5691	+0.0126
128	0.7247	0.6921	+0.0327
136	0.7389	0.7012	+0.0377

Follow-up sweeps found that explicit diversity regularization is not needed for coverage. The strongest distilled configuration was v10 / TokenLearner-SR conv / S=4 / diversity=0 / per_map_nms / nms_dist=2. It nearly ties LookWhere at k=16 and wins at the moderate/high budgets.

k	div0 SoftWhere NMS	LookWhere	Margin
16	0.2591	0.2622	-0.0031
72	0.6077	0.5691	+0.0386
128	0.7495	0.6921	+0.0575
136	0.7605	0.7012	+0.0593

The fovea-count sweep also supports S=4 as the anchor: S=2 has too few foveal slots, while S=8 remains competitive at high k but is weaker than S=4.

5.4 The main distilled head passes sanity checks

The div0/S=4 distilled head remains teacher-aligned and does not collapse into one unusable map. Compared with the div1 head, it improves teacher recall and IoU but allows more overlap among foveal maps, which appears beneficial for coverage.

Metric	div0 head
Teacher top-k recall	0.536
Teacher top-k IoU	0.373
Random top-k recall	0.100
Map overlap	0.463
Plain multi-foveal recall at k=136	0.676
Per-map NMS recall at k=136	0.7605

5.5 Mini end-to-end classifier-signal training strengthens the result

The mini end-to-end run initialized from the div0/S=4 distilled head and trained for 1000 steps with lambda_cls=1 and lambda_map=1. Coverage improved at every tested k, including k=16, which had been the difficult setting for all distilled heads.

k	LookWhere	Distilled div0	Mini E2E CLS	E2E margin
16	0.2622	0.2591	0.2844	+0.0221
72	0.5691	0.6077	0.6413	+0.0723
128	0.6921	0.7495	0.7664	+0.0743
136	0.7012	0.7605	0.7819	+0.0807

The E2E head also passes the map-quality sanity check: teacher recall improves from 0.536 to 0.571, IoU improves from 0.373 to 0.404, and map overlap drops slightly from 0.463 to 0.425. Plain multi-foveal selection now reaches 0.712 object recall, already above the LookWhere single map at 0.701 before NMS is applied.

Metric	Mini E2E CLS head
Teacher top-k recall	0.571
Teacher top-k IoU	0.404
Random top-k recall	0.099
Map overlap	0.425
SoftWhere aggregate recall at k=136	0.609
SoftWhere top-k/S recall at k=136	0.712

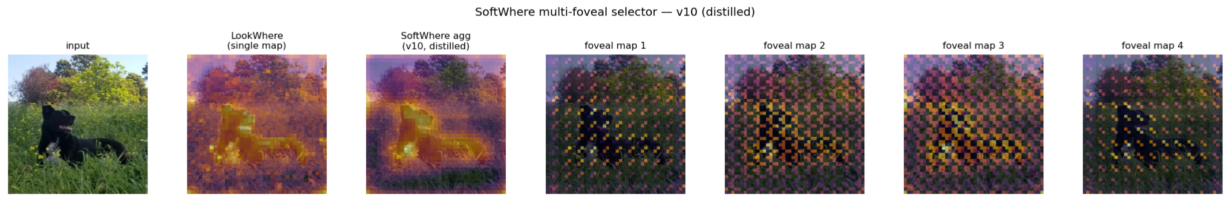

Figure 1. TokenLearner-SR foveal maps. The maps are overlapping teacher-aligned high-resolution scoring maps, not clean semantic object masks; the coverage gain comes from selecting per fovea and applying light NMS.

5.6 Current interpretation

The current best coverage head is softwhere_head_v10_sr_div0_mini_e2e_cls.pt, using v10 / TokenLearner-SR conv / S=4 / diversity=0 / mini E2E CLS / per_map_nms / nms_dist=2. The strongest supported claim is now: TokenLearner-SR plus mini end-to-end classifier-signal training improves ADE20K small-object coverage over LookWhere across all tested patch budgets, k=16, k=72, k=128, and k=136.

The remaining caveat is important: these are object-coverage diagnostics, not the final kNN classification or segmentation protocol. The next research step should test whether the coverage gains transfer to LookWhere's downstream evaluation setting.

6. Proposed Experiments

The immediate next step is downstream evaluation. The coverage diagnostics have already selected the anchor configuration, so the proposal should now spend compute on LookWhere-comparable outcomes instead of more local coverage tuning.

Classification. Run the LookWhere kNN ImageNet-HR class-token protocol at k=16, k=72, and k=128, comparing the stop-gradient LookWhere selector against SoftWhere initialized from the current mini E2E CLS head.

Segmentation. Run the LookWhere ADE20K patch-token kNN segmentation protocol at the same k values. This is the strongest test of whether object-coverage gains translate into denser scene understanding.

Training scale-up. Extend the mini E2E run into the closest feasible version of the LookWhere selector-ablation training setup, retaining map KL regularization and monitoring teacher agreement, IoU, and map overlap throughout.

Ablations. Keep v10 / TokenLearner-SR conv / S=4 / diversity=0 / nms_dist=2 as the anchor, and ablate per-map NMS, aggregate selection, S, map resolution, lambda_cls, lambda_pat, lambda_map, and whether initialization from the distilled head is required.

Coverage. Continue reporting ADE20K small multi-object recall as a mechanism diagnostic alongside downstream kNN metrics, using the same 500-image/341-multi-object evaluation for comparability with the preliminary results.

7. Risks and Mitigations

Risk	Mitigation
Coverage gains do not transfer to kNN classification or segmentation	Treat coverage as a mechanism diagnostic, not the final claim. Run the exact LookWhere kNN protocols and report coverage only as supporting evidence.
The positive result is selection-policy sensitive	Use per-map NMS with nms_dist=2 as the preregistered anchor and include aggregate top-k, per-map top-k, and distance-penalty ablations.
Mini E2E training may not scale to full pretraining	Retain the teacher KL map regularizer, initialize from the distilled div0 head, and monitor teacher agreement, IoU, overlap, and coverage during training.
Foveal maps are misinterpreted as object masks	Describe them as teacher-aligned scoring maps. The method claim is complementary patch selection, not semantic mask discovery.
Very low patch budgets remain hard	The E2E head now beats LookWhere at k=16 in coverage, but downstream k=16 should be treated as the stress test and analyzed separately.

8. Success Criteria

Primary success remains LookWhere-comparable downstream performance: SoftWhere should match or exceed the stop-gradient LookWhere selector on kNN classification at k=16, k=72, and k=128, and on ADE20K kNN segmentation at the same k values. The published LookWhere bars are classification 50.9, 63.0, and 66.5, and segmentation 51.1, 60.2, and 62.9.

Secondary success is a reproducible coverage advantage under the diagnostic harness. The current mini E2E head already clears this bar locally, beating LookWhere object recall by +0.0221 at k=16, +0.0723 at k=72, +0.0743 at k=128, and +0.0807 at k=136. A clean negative downstream result would still be publishable if it explains why coverage improvements fail to transfer.

9. Compute Budget

The completed diagnostics were lightweight: distillation and mini E2E runs finished in minutes to hours on the available GPU setup, and the ADE20K coverage sweeps used 500 validation images with 341 multi-object examples. The next expensive component is the LookWhere-comparable kNN classification and segmentation evaluation, plus any full selector-ablation training needed to make the E2E result fair.

For the full selector ablations, LookWhere used ViT-S and trained for 100 to 400 epochs on ImageNet-1k. A single A100-class GPU can run individual configurations, while multiple GPUs are useful for the sweep. The teacher forward pass remains the major per-step cost, so the practical plan is to first validate the current head in downstream evaluation, then scale only the smallest set of ablations needed to explain the result.

References

A. Fuller et al. *LookWhere? Efficient Visual Recognition by Learning Where to Look and What to See from Self-Supervision.* NeurIPS 2025. arXiv:2505.18051 (Table 12 and Fig. 6–7, selector-training ablations).

M. Ryoo et al. *TokenLearner: What Can 8 Learned Tokens Do for Images and Videos?* NeurIPS 2021. arXiv:2106.11297.